Real Analysis: Exercise 1

Due date:

Question 1. Let $A = \{1, 2, apple\}$, $B = \{2, apple, 1, apple\}$, $C = \{2, 1\}$, $D = \{1, 2, pear\}$.

1. Is $1 \in A$?
2. Is $10 \in A$?
3. Is $A = B$?
4. Is $A = C$?
5. Is $C \subseteq A$?
6. Is $C \subset A$?
7. Is $A \subseteq C$?
8. Is $D \subseteq B$?

Question 2. In this question, the universal set is

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, fish, fowl\}.$$

Let $A = \{1, 2, 3\}$, $B = \{2, 3, fish\}$ and $C = \{2, fowl, 7, 8\}$. Work out each of the following sets using only the definition of union, intersection and complement.

1. $A \cap (B \cup C)$.
2. $(A \cap B) \cup (A \cap C)$.
3. $A \cup (B \cap C)$.
4. $(A \cup B) \cap (A \cup C)$.
5. C^c.
6. $(C^c)^c$.
7. $(A \cap C)^c$.
8. $A^c \cup C^c$.
9. $(A \cup B)^c$.
10. $A^c \cap B^c$.

Question 3. Express each of the following sets as a single interval.

1. $(1, 3) \cup (2, 15)$.
2. $[1, 8] \cap [4, 16]$.
3. $[66, 76] - [72, 100]$.
4. $[0, \infty) \cup (-10, 10)$.
5. $[27, 29] - (26, 28)$

Question 4. If X, Y are sets containing m, n elements respectively, how many elements are there in the Cartesian product $X \times Y$?

Question 5.
1. Find, if possible, sets A and B, each with infinitely many elements and such that $A \cap B = \{0\}$ and $A \cup B = \mathbb{Z}$.
2. Find, if possible, sets C and D such that $C \cup D = \{b, i, g\}$ and $C \cap D = \{s, m, a, l, l\}$.

Question 6.
1. Let A be a set with m elements. Show that the power set $\mathcal{P}(A)$ (the set of all subsets of A) has 2^m elements.
2. Let A and B be subsets of a universal set U. Prove De Morgan's laws:

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$